

Prakash Thermal Power Station (2 x 60 MW)

Machine Master Data Sheet | Digital Twin Reference Dataset | Generated 16-Jul-2026

ID	Machine Name	Zone / Location	Manufacturer	Capacity / Rating	Installed	Status	Last Maint.	Next Maint.
M01	Wagon Tippler & Coal Unloading	Coal Handling Plant	TRF Ltd.	2 x 20T wagons/cycle	2016	Operational	05-Jun-2026	05-Sep-2026
M02	Coal Crusher House	Coal Handling Plant	McNally Bharat	150 TPH, ring granulator	2016	Operational	05-Jun-2026	05-Sep-2026
M03	Coal Conveyor System	Coal Handling Plant	TRF Ltd.	200 TPH, belt width 1000mm	2016	Operational	10-Jun-2026	10-Sep-2026
M04	Stacker-cum-Reclaimer	Coal Yard / Stockpile	TRF Ltd.	Stockpile capacity 15,000T	2016	Operational	12-Jun-2026	12-Sep-2026
M05	Boiler Unit-1 (AFBC)	Boiler House	Thermax	230 TPH steam, 88 kg/cm2	2015	Operational	18-Jun-2026	18-Sep-2026
M06	Boiler Unit-2 (AFBC)	Boiler House	Thermax	230 TPH steam, 88 kg/cm2	2015	Operational	20-Jun-2026	20-Sep-2026
M07	ID / FD / PA Fan Set	Boiler House	Kirloskar	3 x fans, 750 kW total	2015	Under Maintenance	01-Jul-2026	24-Jul-2026
M08	Steam Turbine-Generator Unit-1	Turbine Hall	BHEL	60 MW, 3000 RPM, 11kV	2015	Operational	22-Jun-2026	22-Sep-2026
M09	Steam Turbine-Generator Unit-2	Turbine Hall	BHEL	60 MW, 3000 RPM, 11kV	2015	Operational	24-Jun-2026	24-Sep-2026
M10	Surface Condenser & Vacuum Pump	Turbine Hall	BHEL	Vacuum -0.9 kg/cm2	2015	Operational	24-Jun-2026	24-Sep-2026
M11	Induced Draft Cooling Tower	Cooling Tower	Paharpur	2-cell, 9000 m3/hr	2015	Operational	15-Jun-2026	15-Sep-2026
M12	DM Water Treatment Plant	DM Water Treatment	Ion Exchange India	40 m3/hr, 2-bed DM	2016	Operational	08-Jun-2026	08-Sep-2026
M13	Electrostatic Precipitator (ESP)	ESP & Chimney	Thermax	99.5% collection efficiency	2015	Operational	28-Jun-2026	28-Sep-2026
M14	Chimney / Stack	ESP & Chimney	In-house (RCC)	Height 120m, dual flue	2015	Operational	N/A (annual inspection)	15-Dec-2026
M15	Ash Handling Plant	Ash Handling & Ash Pond	McNally Bharat	Dry ash, 40 TPH capacity	2016	Operational	10-Jun-2026	10-Sep-2026
M16	Generator Transformers & Switchyard	Switchyard / Substation	BHEL / Crompton Greaves	2 x 65 MVA, 11/220kV	2015	Operational	16-Jun-2026	16-Sep-2026
M17	Raw Water Pump House	Raw Water Reservoir	Kirloskar Brothers	3 x pumps, 500 m3/hr each	2016	Operational	14-Jun-2026	14-Sep-2026

Notes: Machine IDs correspond to markers on the plant floor plan drawing. Maintenance intervals are quarterly preventive maintenance (PM) cycles unless otherwise flagged. This dataset is synthetic sample data generated for digital twin development and testing purposes.